



A deep dive into the RecQ interactome: something old and something new

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Abstract

RecQ family helicases are found in all domains of life and play roles in multiple processes that underpin genomic integrity. As such, they are often referred to as guardians or caretakers of the genome. Despite their importance, however, there is still much we do not know about their basic functions *in vivo*, nor do we fully understand how they interact in organisms that encode more than one RecQ family member. We recently took a multi-omics approach to better understand the *Saccharomyces cerevisiae* Hrq1 helicase and its interaction with Sgs1, with these enzymes being the functional homologs of the disease-linked RECQL4 and BLM helicases, respectively. Using synthetic genetic array analyses, immuno-precipitation coupled to mass spectrometry, and RNA-seq, we found that Hrq1 and Sgs1 likely participate in many pathways outside of the canonical DNA recombination and repair functions for which they are already known. For instance, connections to transcription, ribosome biogenesis, and chromatin/chromosome organization were uncovered. These recent results are briefly detailed with respect to current knowledge in the field, and possible follow-up experiments are suggested. In this way, we hope to gain a wholistic understanding of these RecQ helicases and how their mutation leads to genomic instability.

Keywords DNA helicase · RecQ · Hrq1 · Sgs1 · RECQL4 · BLM

Introduction

DNA helicases represent one of the only classes of enzymes involved in nearly all facets of genome maintenance (Bochman 2014). These proteins couple the energy of ATP hydrolysis to nucleic acid binding and translocation to unwind DNA, RNA–DNA hybrids, and non-canonical secondary structures, which facilitates multiple aspects of DNA replication, recombination, and repair, telomere maintenance, and transcription, among other processes (Bochman et al. 2012; Matson et al. 1994). As vital pieces of the genome integrity machinery, DNA helicases are encoded by the genomes of all organisms, are often evolutionarily

conserved, and helicase mutations are linked to or causative of diseases in humans. The RecQ proteins belong to one of the most extensively studied families of conserved helicases. To date, all of the characterized RecQs display DNA-dependent ATPase activity and 3′–5′ helicase activity (Bernstein et al. 2010), though their preferred substrates differ between family members and between functional homologs depending on the species of origin (Croteau et al. 2014; Rogers et al. 2017).

The founding member of this family, the *Escherichia coli* RecQ, was identified as a helicase in 1990 (Umezumi et al. 1990) and is generally thought to be the only RecQ found in prokaryotes. However, nature is not that simple. The genomes of some species of bacteria appear to be devoid of RecQ helicases altogether (Bochman et al. 2011), while others contain multiple family members. For instance, recent work indicates that pathogenic strains of *E. coli* often encode a second RecQ family helicase named RqlH, and RqlH homologs are found in a diverse array of other bacterial species (Russell and Mulvey 2015). Similarly, archaeal species are considered to lack RecQ helicases, having found functional replacements in enzymes like Hjm that are structurally and phylogenetically distinct from *E. coli* RecQ but display

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RecQ-like biochemical activities (*e.g.*, Holliday junction migration) (Fujikane et al. 2006; Oyama et al. 2009). Again, this is not actually the case though. Many prokaryotes, archaea included, encode an enzyme called SfhH (Yakovleva and Shuman 2012) or MrfA (Burby and Simmons 2019) that is homologous to the Hrq1 and RECQL4 eukaryotic RecQ family helicases, which we collectively refer to as the RecQ4 subfamily helicases.

The RecQ confusion extends to eukaryotes as well. Many researchers consider Sgs1, a homolog of the human BLM helicase (Cejka and Kowalczykowski 2010; Gravel et al. 2008; Heo et al. 1999; Lillard-Wetherell et al. 2005), to be the only RecQ found in the budding yeast *Saccharomyces cerevisiae*. However, bioinformatics, biochemistry, genetics, and structural biology demonstrate that the *S. cerevisiae* Hrq1 helicase is a human RECQL4 homolog (Barea et al. 2008; Bochman et al. 2014; Nickens et al. 2018; Rogers and Bochman 2017; Rogers et al. 2017, 2020a). The Y' subtelomeric repeats in *S. cerevisiae* also contain the *YRF1* genes, which encode Y'-Help1 helicases that are expressed during telomerase-independent telomere maintenance (Yamada et al. 1998). In budding yeast and other simple eukaryotes, Y'-Help1 and its homologs are suggested to be RecQ family helicases with important roles in recombination-mediated telomere lengthening (Hansen et al. 2006; Inglis et al. 2005; Yamada et al. 1998), though this family of enzymes is woefully understudied at present.

In contrast to the chaotic world of RecQs among microbes, it is well established that the human genome encodes five RecQ helicase family members (RECQL1,

WRN, BLM, RECQL4, and RECQL5) that play various roles in genome maintenance (Croteau et al. 2014). Mutations in three of these enzymes are causative of disease: BLM/Bloom syndrome, WRN/Werner syndrome, and RECQL4/Baller-Gerold, RAPADILINO, and Rothmund-Thomson syndromes (Hanada and Hickson 2007; Monnat 2010; Opresko et al. 2004; van Brabant et al. 2000). While RECQL1 and RECQL5 mutations are not yet associated with any diseases, they do predispose individuals to cancer (Debnath and Sharma 2020; Tavera-Tapia et al. 2019). Despite the obvious importance of these helicases to genome integrity, we largely do not know all of their typical *in vivo* functions nor what goes wrong in RecQ mutant cells that ultimately results in a disease state.

The above conundrum is what has been driving our own research into the roles of RecQ helicases in genome maintenance. Over the past decade, we have been studying *S. cerevisiae* Hrq1 *in vitro* and *in vivo* as a simple model to learn more about RECQL4 biology. To date, we have reported the basic biochemical properties of both enzymes (Rogers et al. 2017), begun teasing apart the mechanism of Hrq1 and RECQL4 in Fanconi anemia-independent DNA inter-strand crosslink (ICL) repair (Bochman et al. 2014; Rogers et al. 2020a, c), and demonstrated a synergistic interaction between Hrq1 and the Pif1 helicase that regulates telomerase activity and telomere length homeostasis (Bochman et al. 2014; Nickens et al. 2018, 2019) (Fig. 1). However, these functions only scratch the surface of Hrq1 biology. Moving forward, we sought to define the physical and genetic interactomes of Hrq1 in yeast, as well as to investigate

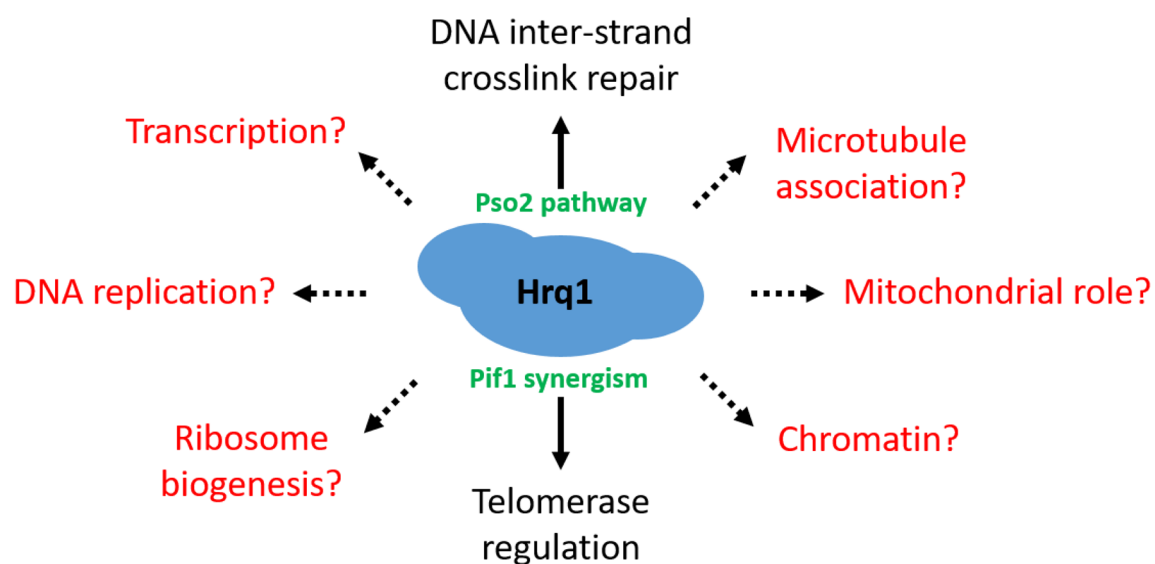


Fig. 1 The known and putative genome maintenance roles of Hrq1. Hrq1 stimulates the translesional activity of the Pso2 nuclease during DNA ICL repair (Rogers et al. 2020a), and it synergizes with the Pif1 helicase to regulate telomerase activity at telomeres and double-

strand breaks (Bochman et al. 2014; Nickens et al. 2018). Our recent data also suggest roles in the processes listed in red (Rogers et al. 2020b; Sanders et al. 2020)

the crosstalk between Hrq1 and Sgs1 in vivo. Our recent efforts (Rogers et al. 2020b; Sanders et al. 2020) are briefly reviewed below and discussed with an eye toward outstanding questions in the RecQ field.

Limitations of genetic screens and a focus on inactive helicase mutants

Our extensive experience with *S. cerevisiae* Hrq1 biology, combined with the broad suite of known functions of human RecQ helicases, prompted us to take a comprehensive, multi-omics approach to expand our understanding of processes involving Hrq1 (e.g., ICL repair) and to identify novel functions. Utilizing the powerful genetic screening tools available in budding yeast, we performed synthetic genetic array (SGA) analysis using deletion (*hrq1Δ*) and catalytically inactive (*hrq1-K318A*) mutants crossed to the yeast single-gene deletion (Giaever et al. 2002) and temperature-sensitive allele (Kofoed et al. 2015) collections (Rogers et al. 2020b). Knowing that human BLM and RECQL4 function synergistically (Singh et al. 2012), we also performed SGA analyses using all possible combinations of deletion and inactive mutants of their *S. cerevisiae* homologs Sgs1 and Hrq1: *hrq1Δ*, *sgs1Δ*, *hrq1-K318A*, *sgs1-K706A*, *hrq1Δ sgs1Δ*, *hrq1Δ sgs1-K706A*, *hrq1-K318A sgs1Δ*, and *hrq1-K318A sgs1-K706A* (Sanders et al. 2020). To complement this genetic approach, we utilized proteomics to define the physical interactome of Hrq1, alongside transcriptomics to understand how cells respond to the absence of Hrq1 activity (Rogers et al. 2020b). Herein, we briefly review our approaches and findings, as well as discuss how we and others can leverage these data to continue to learn about how RecQ helicases support genome integrity.

The benefits of our approach

Previously performed genetic screens have provided incredibly valuable and extensive datasets that are useful resources to expand our understanding of how genes interact and identify new avenues of research (Costanzo et al. 2010, 2016). However, when a narrow range of genetic targets (i.e., RecQ helicases) is selected for study and deletion alleles have been the focus of previous work, it is imperative to repeat the genetic screens for those targets, as some interactions are potentially missing from global screens. Additional mutant background strains should also be considered to observe interactions that are absent in strains completely lacking certain genes. We chose Walker A box lysine-to-alanine substitution mutants in Hrq1 and Sgs1 because these mutations ablate ATPase activity, creating “dead” helicase alleles. Inactive DNA helicase mutants have been shown to act as dominant negatives, which often better model diseased

phenotypes than gene deletions (Wu and Brosh 2010). Mutants that disrupt activity but still express protein can also provide biological insights, as these mutants may be sufficient for some activities while abrogating others (Bochman et al. 2014). Therefore, by screening both the deletion and catalytically inactivate mutants of *HRQ1* and *SGS1*, as well as all possible double mutant combinations of these alleles, we generated a nearly comprehensive list of genetic interactions with the genes encoding RecQ helicases in yeast.

To accompany the genomic analyses described above, we also performed immunoprecipitation-mass spectrometry (IP-MS) of Hrq1-3xFLAG to identify its in vivo interaction partners. Similarly, we used RNA-seq of wild-type, *hrq1Δ*, and *hrq1-K318A* strains to determine changes in gene expression in the absence of Hrq1 activity, with the impetus for this latter experiment coming from SGA data suggesting a link between *HRQ1* mutations and transcription defects (Rogers et al. 2020b). Our hope was that this multi-omics approach would generate overlapping and supporting data, underscoring the important role of Hrq1 in known pathways and uncovering new leads. While this was partially successful, the genomics, proteomics, and transcriptomics datasets were also quite distinct (see below).

The pitfalls of our approach

Although our experiments provided valuable information regarding the participation of RecQ helicases in known and unknown pathways, there are limitations inherent to such high-throughput approaches. For example, our SGA analyses hits were based on cell growth, which is a narrow, albeit important, readout. However, these data do not speak to other phenotypes such as changes to cell morphology, protein subcellular localization, or the transcription of genes not involved in general cell growth. Furthermore, this screen was performed using complete media without perturbation of any cellular processes. Finally, the parameters impacting cell growth kinetics (i.e., lag phase, doubling rate, and maximum expansion) stochastically vary (Ngo et al. 2019). Even with individual SGA crosses performed in quadruplicate, this variation can lead to false positives, false negatives, averaged results that dampen real effects below the significance threshold, and overlooking effects that may present in liquid culture growth instead of growth on a solid medium.

Thus, our dataset is likely missing some, at least subtle interactions. Certainly, it lacks enrichment of genetic or physical interactions that are induced by DNA damage, which is infrequent in the absence of exogenous sources, especially with respect to ICLs (Rogers et al. 2020d). As stated above, data analysis techniques are also a concern for genome- and proteomewide studies; statistically determined data cutoffs have the potential to generate false negatives and false positives. For example, *hrq1Δ* mutant cells were within

a dataset of cisplatin-sensitizing *S. cerevisiae* gene deletions but fell below the original statistical significance cutoff (Lee et al. 2005). The sensitivity of *hrq1Δ* mutant cells to cisplatin was only later validated through alternative analyses and confirmed with cisplatin-dosed spot assays (Bochman et al. 2014; Dittmar et al. 2013). Finally, not all essential genes in the *S. cerevisiae* genome are represented by alleles in the two libraries that we screened, and other potentially informative mutations (*e.g.*, separation of function) in *HRQ1* and *SGS1* were not generated for inclusion in the SGA analyses. Thus, we can only term the double and triple mutant screens as “nearly comprehensive” because it is the most thorough performed to date. The inclusion of the IP-MS and RNA-seq experiments was an attempt to fill in some of the blanks left by the SGA approach.

One obvious follow-up to our recent work would be to include agents that induce DNA damage or inhibit transcription. Coupling SGA with chemical perturbations has been a highly successful chemogenomics approach (Lee et al. 2014) and can yield insight into specific pathways if the type of chemical damage is specific, *e.g.*, creating ICLs vs. double-strand breaks. Likewise, treatment with various chemicals could enrich for distinct subsets of Hrq1 interaction partners, yielding proteomics data that could discretely identify pathways and/or multi-protein complexes that include Hrq1 *in vivo*.

The *S. cerevisiae* *hrq1* interactome

Genomics

SGA analyses of the eight different *hrq1* and/or *sgs1* mutant strains revealed hundreds of genetic interactions (positive and negative) per strain. Of particular interest is that the double mutant strains displayed nearly twice the additive number of genetic interactions as the respective single mutant strains (Sanders et al. 2020). These sets of synergistic interactions are generally enriched for the gene ontology (GO) terms of DNA repair, DNA replication, transcription, and other processes related to genome integrity.

The *hrq1Δ* and *hrq1-K318A* interactions can be specifically divided into ten main GO-term categories each for positive and negative genetic interactors. The top ten negative GO-term categories (listed from highest to lowest frequency) are transcription by RNA polymerase II, regulation of organelle organization, DNA repair, chromatin organization, mitotic cell cycle, peptidyl-amino acid modification, cytoskeleton organization, mitochondrion organization, organelle fission, and response to chemical. Similarly, the top ten positive GO term categories (listed from highest to lowest frequency) are mitotic cell cycle, cytoskeleton organization, regulation of organelle organization, lipid metabolic

process, DNA repair, transcription by RNA polymerase II, chromatin organization, chromosome segregation, organelle fission, and rRNA processing (Rogers et al. 2020b). Surprisingly, although interactions appear that reflect the known roles of Hrq1 and RECQL4 in DNA repair, telomere maintenance, and in the mitochondria (Bochman et al. 2014; Croteau et al. 2012; Ghosh et al. 2011; Koh et al. 2015; Nickens et al. 2018; Rogers et al. 2017; Singh et al. 2012), these functions are under-represented by the GO terms compared to processes not previously linked to Hrq1 or RECQL4 activity, likely reflecting the conditions under which the analysis was performed in (*e.g.*, no DNA damaging agents) as discussed above.

Proteomics

IP-MS of Hrq1-3xFLAG revealed 77 significant interacting partners. These proteins are enriched for many of the same GO terms seen in the SGA screens: rRNA processing, transcription by RNA polymerase II, DNA repair, chromatin organization, and peptidyl-amino acid modification (Rogers et al. 2020b). However, missing from these results is the Pso2 nuclease that is physically stimulated by Hrq1 to digest through ICLs *in vitro* (Rogers et al. 2020a). Expression of Pso2 is normally very low but increases three- to fourfold under DNA damaging conditions (Wolter et al. 1996). This again highlights the importance of following up these results with screens under growth conditions that affect specific processes, such as treating cells with an ICL-inducing agent to enrich for proteins in the Hrq1/Pso2 ICL repair pathway (Rogers et al. 2020c).

Transcriptomics

Because transcription GO terms were enriched in both the SGA and IP-MS analyses with respect to Hrq1, we were interested in the potential effect of Hrq1 mutation on the global transcriptome in yeast. When cells were grown in the presence of the general transcription inhibitor caffeine (Kuranda et al. 2006), *hrq1Δ* and *hrq1-K318A* strains displayed significant sensitivity compared to wild type. From the RNA-seq results, we found significant changes in the regulation of 107 (down) and 28 (up) genes in *hrq1Δ* cells, and 301 (down) and 124 (up) genes in *hrq1-K318A* cells. GO-term enrichment of these differentially expressed genes once again reflects some of the processes implicated by SGA and IP-MS analyses, such as mitotic cell cycle, rRNA processing, and chromosome segregation (Rogers et al. 2020b). Additionally, the most differentially expressed genes included a protein required for survival in cells lacking mitochondrial DNA (*ICY1*) (Dunn and Jensen 2003) and three proteins of unknown function that are induced by ICL damage (*YLR297W*, *TDA6*, and *FMP48*) (Dardalhon et al. 2007).

A plethora of data—where do we go next?

Although the experiments described above yielded both expected and unexpected results, the question remains: how do we and others best utilize these complex datasets of hundreds of interactions to design follow-up experiments? The obvious first steps are to expand the network of interactions in known pathways and identify new functions, such as in the case of Hrq1's involvement in ICL repair and transcription, respectively. Indeed, of particular interest to us is the involvement of RecQ helicases in ICL repair and telomere maintenance. Aside from repeating the SGA, Hrq1 IP-MS, and/or RNA-seq experiments in *S. cerevisiae* cells challenged with ICL-inducing agents, we are particularly interested in the proteins encoded by the ICL-inducible *YLR297W*, *TDA6*, and *FMP48* genes.

Ylr297w is a small (~ 14 kDa) protein that relocates from the nucleus to the vacuole in the presence of replication stress (Giaever et al. 2002; Huh et al. 2003). Tda6 is a 54-kDa protein that has also been found in the vacuole, and deletion of *TDA6* renders cells sensitive to the camptothecin-mimetic *top1-T722A* allele (Reid et al. 2011). Fmp48 (43 kDa) is predicted to be a mitochondrial protein kinase (<https://www.yeastgenome.org/locus/S000003284>). Although the expression of all of these proteins is induced by ICL damage from 8-methoxypsoralen and UVA irradiation, there is no indication of what they may do, if anything, in ICL repair. However, Hrq1 and Pso2 can localize to the mitochondria (Koh et al. 2015), and both are phosphorylated (Bochman, unpublished; Munari et al. 2014)), suggesting a possible link to Fmp48. Single and double deletion strains of *YLR297W*, *TDA6*, and *FMP48* should be constructed to assess their sensitivity to ICL damage, and double mutants with *HRQ1* or *PSO2* should be used for epistasis analysis to determine if they function in the same pathway.

Aside from following up on some of these more obvious leads, we are also interested in pursuing novel functions that Hrq1 may have in vivo. Deciding which hits are real and biologically relevant is an important consideration when using data from high-throughput screens to identify novel functions. For instance, Hrq1 is best known for its role in genome maintenance (Rogers et al. 2020c), but DNA damage was not the top hit in GO term analyses of Hrq1 genetic and physical interactors (Rogers et al. 2020b). Surprisingly, important factors involved in transcription by RNA polymerases I and II were among the most consistent and significant hits across each of our screens. Prior to this work, there was no indication of Hrq1 function in transcription, though it is established that human RECQL5 is involved in transcription by RNA

polymerase II (Hamadeh and Lansdorp 2020). It is tempting to speculate that Hrq1 may function as the *S. cerevisiae* RECQL5 homolog in transcription, as both are linked to transcription elongation (Rogers et al. 2020b; Saponaro et al. 2014). Alternatively, the human homolog of Hrq1, RECQL4, may have a conserved yet uncharacterized role in transcription. Regardless, our decision to directly test for a transcription role of Hrq1 both reinforced that the data from our screens can be applied to biological functions and highlights the value of an open-minded approach to results from high-throughput datasets.

Similarly, GO term analysis identified genes involved in cytoskeleton organization as highly enriched positive and negative genetic interactors (Sanders et al. 2020). RECQL4 was recently identified as a microtubule-associated protein (Yokoyama et al. 2019), suggesting the conservation of a previously unexplored RecQ4 subfamily function. Unfortunately, however, when we attempted to test this hypothesis using an in vitro microtubule co-sedimentation assay (Walker et al. 2019) with recombinant Hrq1, we encountered technical difficulties that will require significant troubleshooting (Rogers et al. 2020b). In any event, this represents another potentially conserved function that can be explored.

Although we focus on Hrq1 here, our SGA data also cover a large swath of Sgs1 biology, and interested readers are encouraged to delve into our recent results (Sanders et al. 2020). Surely, these data can be leveraged by our colleagues who focus on Sgs1/BLM, and we are interested in a potential interaction between Hrq1 and Sgs1 as demonstrated for RECQL4 and BLM (Singh et al. 2012). What else do Hrq1, Sgs1, RECQL4, and BLM do that underpins genomic integrity? Our results suggest putative roles in many pathways, including rRNA processing, ribosome biogenesis, chromatin organization, Okazaki fragment maturation, and many others (Rogers et al. 2020b; Sanders et al. 2020) (Fig. 1). It will be exciting to see how these links are fleshed out, directly and indirectly, as research in the genome integrity field continues. Similarly, deletions of many *S. cerevisiae* genes of unknown function were recovered as positive or negative genetic interactors with our *hrq1* and *sgs1* mutants. Following up on these hits will prove doubly satisfying by revealing more about the roles of RecQ helicase in vivo and assigning names and functions to these uncharacterized factors.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Consent for publication All authors have read and approved the contents of this manuscript.

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